

Recognising individual bats by face: what the multi-seed results actually show

Master’s thesis — results interpretation

Open-set individual recognition • 2 species • 3 model families • 3 backgrounds • 5 seeds/cell
(270 runs)

In one paragraph. Across a re-run of the full experiment matrix, individual bat faces are recognisable well above chance in the favourable conditions (best cells ROC-AUC $\approx$ 0.85–0.89; permutation p from < 0.001 to 0.007 on the thresholded operating point), and the signal is genuinely in the *foreground* — performance survives removing the background entirely. Two limits are equally clear: the models are not background-invariant (randomised backgrounds drop them to near chance), and higher input resolution does not help (224→320 px, mean Δ ROC-AUC = -0.033). Metric-learning embeddings (ArcFace/AdaFace) are more robust than the contrastive Siamese network, especially where training identities are few. A control experiment adds a decisive qualifier: much of the apparent recognition — nearly **all** of it for *Rousettus* — is reproducible from the crop **silhouette** alone, so the signal is as much head-shape as facial texture. The dominant caveat throughout is the tiny open-set test (two held-out identities per species), so every claim is directional, not definitive.

1 What the experiment measured

The task is **open-set** individual recognition: the identities used to test the model were never seen in training. Each species’ data is split by identity, and the held-out *test* set is small (Table 1) — the single most important fact for interpreting everything below.

Table 1. Identity-disjoint splits.

Species	Train id.	Val id.	Test id.	Train img	Test img
Mauritius	12	2	2	373	77
Rousettus	8	2	2	619	137

Three model families were compared: a **Siamese** 4-conv contrastive network (learns a similarity from pairs) and two margin-based metric-learning heads on a ResNet-50 backbone, **ArcFace** and **AdaFace** (learn an identity-structured embedding). Each was trained on three **background variants** of the same face crops — *original* (natural background kept), *green* (face segmented onto a uniform field, i.e. background removed), and *random* (face composited onto varied, identity-unrelated backgrounds) — at five seeds each. Verification quality is ROC-AUC; identification is top-1/mAP; a **permutation test** checks each result against a shuffled-label null.

How to read the numbers. With only two test identities, a single run is noisy: one pairing can swing ROC-AUC by > 0.1 . Every cell is therefore a **mean $\pm$ standard deviation over five seeds**, and chance is ROC-AUC = 0.5. Differences smaller than roughly one standard deviation are ties. The permutation test asks the stricter question — could this arise from a model that learned nothing? — and for the strong cells the answer is no (p from < 0.001 to 0.007). These p-values are computed on the thresholded operating point (F1 and accuracy/precision/recall), not on ROC-AUC itself, over $B = 1000$ label shuffles. The smallest value B can resolve is

$1/(B+1) \approx 0.001$, so cells reported at that value are *censored* at the floor rather than estimated: the truth lies at or below it.

2 The headline results

Table 2. Test ROC-AUC (mean $\pm$ std, 5 seeds), edge 112 px. **Green** ≥ 0.75 , **amber** 0.60–0.75, **red** < 0.60 (approaching chance).

Model	Background	Mauritius	Rousettus
Siamese	original	0.880 $\pm$.059	0.614 $\pm$.106
Siamese	green	0.798 $\pm$.105	0.421 $\pm$.051
Siamese	random	0.700 $\pm$.050	0.454 $\pm$.035
ArcFace	original	0.885 $\pm$.068	0.740 $\pm$.119
ArcFace	green	0.777 $\pm$.122	0.774 $\pm$.085
ArcFace	random	0.611 $\pm$.073	0.508 $\pm$.135
AdaFace	original	0.845 $\pm$.071	0.739 $\pm$.056
AdaFace	green	0.809 $\pm$.114	0.757 $\pm$.127
AdaFace	random	0.592 $\pm$.061	0.551 $\pm$.102

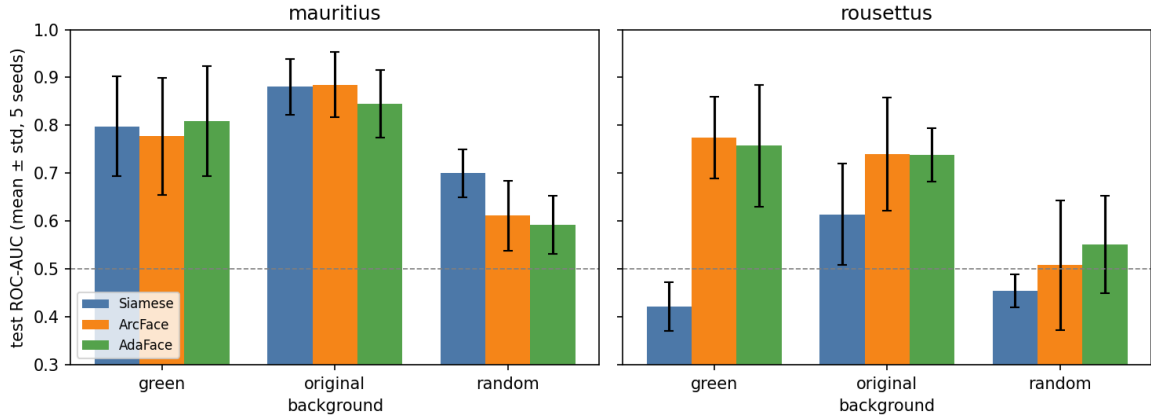


Figure 1. Test ROC-AUC (mean $\pm$ std) by background, per species; dashed line is chance (0.5). Two patterns recur: *random* backgrounds are lowest everywhere, and *Rousettus* is both lower and more variable than *Mauritius*.

3 Finding 1 — the recognition is real

The concern with a two-identity test is that a good score is a fluke. The permutation test re-scores each model against 1000 label shuffles; for the strong cells (*Mauritius* ArcFace/AdaFace on original background) the observed operating-point score beats *every* shuffle, giving the smallest p-value $B = 1000$ can express ($p < 0.001$); the weaker of the strong cells reach $p = 0.007$.

Defensible claim. In the favourable conditions the models discriminate individuals at a rate shuffled-label chance does not reproduce — the learned representation carries genuine identity information.

4 Finding 2 — foreground, not background (but not background-invariant)

If the high scores were the model memorising the enclosure rather than the bat, removing the background should destroy performance. It does not:

- **Background removed (green):** embedding models still recognise well — ArcFace 0.78 / 0.77 and AdaFace 0.81 / 0.76 (Mauritius / Rousettus), on par with the natural background.
- **Background randomised (random):** the same models collapse toward chance (0.51–0.61).

Green working while random fails is the tell: a clean uniform field lets the network read the foreground; arbitrary cluttered backgrounds act as distractors it was never taught to suppress. So the model has learned a foreground representation that is *not robust* to background variation. *Which part* of the foreground it uses — the face, or the crop shape — is the subject of Section 7.

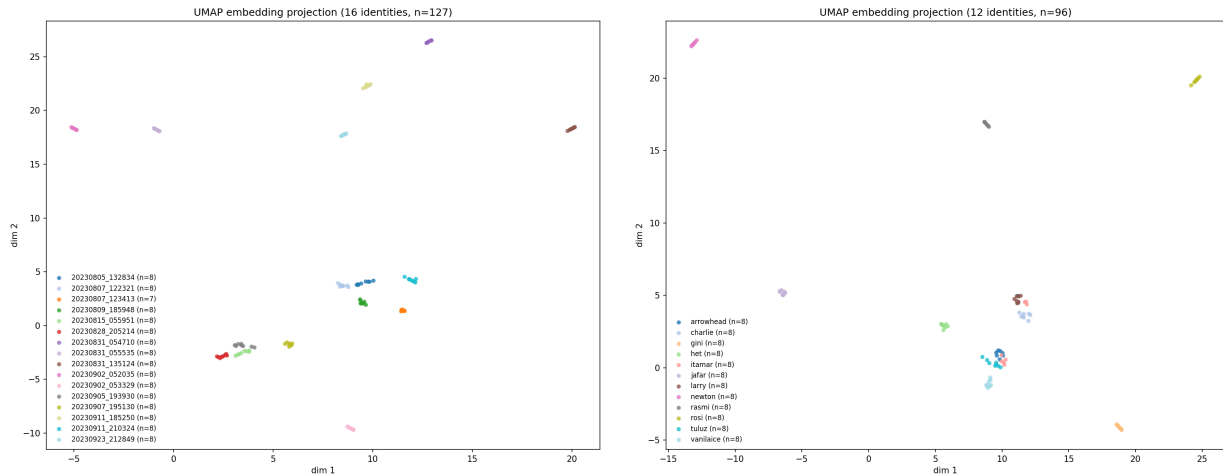


Figure 2. UMAP of ArcFace test embeddings, coloured by identity (left Mauritius, right Rousettus, original background). Tight, separated colour clusters mean same-individual images sit together and different individuals are pushed apart — the geometric basis of the ROC-AUC numbers.

5 Finding 3 — embeddings beat Siamese, most where identities are scarce

The families diverge sharply on Rousettus (fewer training identities, 8 vs 12): the Siamese network barely functions (0.61 original, **0.42 green — below chance**), while ArcFace/AdaFace hold 0.74–0.77. The Mauritius→Rousettus drop is therefore largest for Siamese (green: 0.80→0.42). The contrastive objective learns a metric from pairs and needs a diversity of identities to generalise; with only eight it underfits. Margin-based heads extract more per identity and transfer better to the open-set test.

6 Finding 4 — higher resolution does not help

Retraining the embedding models on genuinely higher-resolution crops at 224 and 320 px did not improve recognition: mean change 224→320 was -0.033 ROC-AUC (green -0.048 , original -0.036 , random -0.015). GradCAM localises the eyes *more* sharply at higher resolution (Fig. 4), so the model is not starved of detail — it already has what it needs at 112 px.

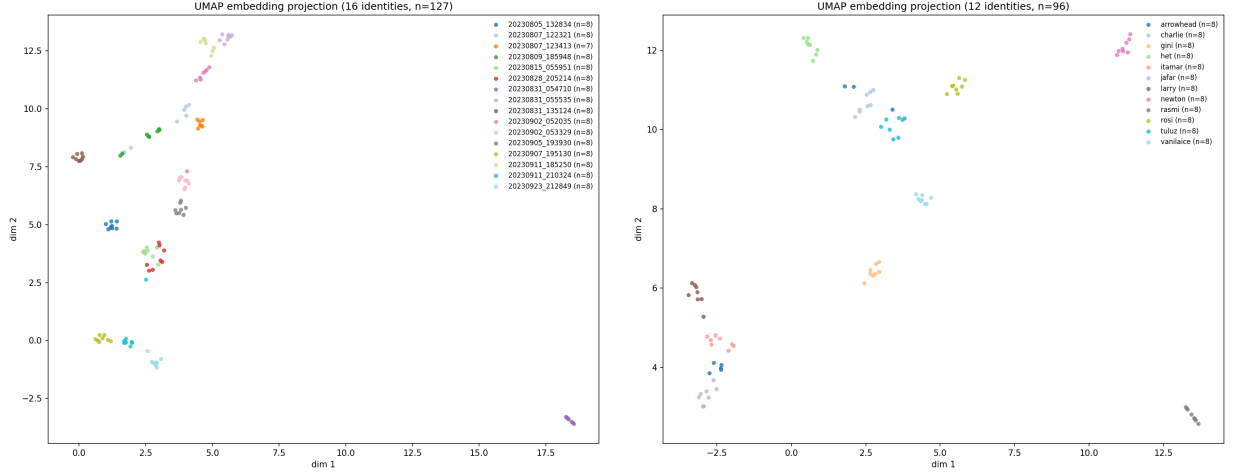


Figure 3. Siamese embeddings, UMAP (left Mauritius, right Rousettus, original) — the family with the largest species gap. Mauritius shows structure; Rousettus is diffuse, the geometric counterpart of its near-chance ROC-AUC. (This is why the resolution study excludes Siamese: its 4-conv net is fixed at 105 px.)

7 Internal validity: is it the face, or the crop shape?

Two terms, used throughout this section.

Shape (silhouette) — the outline of the segmented head crop only: head size, ear notches, jaw contour. The black-vs-white boundary in the “Silhouette” panels (Figs. 5–6), with all interior detail erased.

Texture — shorthand for everything *inside* that outline: the eyes (position, spacing, iris), the snout and muzzle patterning, fur colour and markings, shading — the internal facial appearance a person would use to tell two bats apart. (“Texture” is the standard computer-vision term; it means internal appearance here, not just fine fur detail.)

Finding 2 showed recognition survives removing the background (green). But green keeps the tight segmentation **silhouette** — the head/face outline — which is itself identity-correlated (the “Silhouette” panels: head size, ear notches and jaw contour visibly differ between individuals). So “green works” might reflect the model reading the *shape* of the crop, not the *texture* of the face. Two controls test this directly.

Control 1 — train on shape alone (silhouette)

Retrain the full matrix on the pure silhouette, with identical identities and splits, and compare to green (shape + texture):

Table 3. Silhouette (shape only) vs green, test ROC-AUC (mean $\pm$ std, 5 seeds). Red Δ = silhouette matches/beats green (shape dominates); green Δ = texture adds value.

Species	Model	Silhouette	Green	Δ (silh–green)
Mauritius	Siamese	0.697 $\pm$.023	0.798	–0.101
Mauritius	ArcFace	0.649 $\pm$.086	0.777	–0.128
Mauritius	AdaFace	0.611 $\pm$.130	0.809	–0.198
Rousettus	Siamese	0.698 $\pm$.032	0.421	+0.277
Rousettus	ArcFace	0.823 $\pm$.038	0.774	+0.049
Rousettus	AdaFace	0.820 $\pm$.075	0.757	+0.063

Shape alone is **well above chance everywhere** (0.61–0.82). For Mauritius the silhouette runs 0.10–0.20 *below* green, so internal texture genuinely adds value. For Rousettus the silhouette **matches or beats** green — Siamese by +0.28 — so the bare outline is as discriminative as the textured face, or more.

Control 2 — occlude one cue on the real-face model (5 seeds)

Table 4. Occluding one cue in the test set of a model trained on real faces (ArcFace/original, mean over 5 seeds). Bigger drop = stronger reliance on the removed cue.

Species	Test-set ablation	ROC-AUC	Δ vs clean
Rousettus	clean	$0.793 \pm .135$	—
Rousettus	shape removed (texture only)	$0.672 \pm .117$	−0.122
Rousettus	texture removed (shape only)	$0.774 \pm .141$	−0.019
Mauritius	clean	$0.808 \pm .185$	—
Mauritius	shape removed (texture only)	$0.884 \pm .047$	+0.076 (noise)
Mauritius	texture removed (shape only)	$0.815 \pm .101$	+0.007 (noise)

For Rousettus, removing the *shape* costs 6× more than removing the *texture* (−0.122 vs −0.019) — it relies on shape, corroborating Control 1. For Mauritius the deltas are swamped by the two-identity test’s ± 0.20 noise and are inconclusive (occlusion even nominally “improves” it — a sign of pure variance).

Why this is not negligible

- **It is most of the signal, not a small correction.** For Rousettus a binary silhouette with zero facial detail reaches ROC-AUC 0.82 (ArcFace) — *higher* than the textured face (0.77). Nearly all of the reported Rousettus recognition is reproducible from the outline alone.
- **Two independent controls agree.** For Rousettus, silhouette $\geq$ green (Control 1) *and* shape-removal hurts most (Control 2).
- **It changes what the result means.** A “face recognition” score that survives deleting the face — keeping only the crop outline — is measuring head/crop *geometry*, not facial identity. The silhouette is the segmentation mask, whose contour (head size, ear shape, jaw line) is stable within an individual and variable across individuals: a per-individual shortcut a CNN will exploit.
- **It is species-specific.** For Mauritius texture *does* contribute (silhouette drops below green), so the face claim is partly supported — though shape still accounts for a large share. For Rousettus a distinct facial-texture contribution is not established by these data.

State it carefully. Report the silhouette baseline next to every headline number. Honest phrasing: Mauritius — “individuals are recognised from face crops, with contributions from both facial texture and head-crop shape”; Rousettus — “individuals are separable largely by head-crop **shape**; a distinct facial-texture contribution is not demonstrated.” Isolating texture cleanly would require shape-normalised inputs (a fixed aperture) at training time, not just at test time.

8 What the thesis learns

- **Feasibility.** Automated individual identification of these fruit-bat species from face crops is possible above chance — a proof of concept, not yet a deployable tool.
- **Architecture.** Prefer margin-based embedding learning (ArcFace/AdaFace) over contrastive Siamese, especially with few identities.
- **The real bottleneck is data.** Neither model capacity nor input resolution is limiting; the scarce resource is the *number of individuals*.
- **Background must be controlled, and shape is a confound, not a footnote** — a silhouette baseline recognises individuals on its own, so the “face” framing must be qualified and the silhouette number reported beside every accuracy.

9 Limitations to state up front

- **Two-identity open-set test** — wide confidence intervals; conclusions are about patterns across the matrix, not any single cell.
- **Background robustness gap** — the random-background collapse means the model would not transfer to unfamiliar backgrounds without further work.
- **Shape confound** — a large share of performance (nearly all for *Rousettus*) is reproducible from the crop silhouette; isolating texture needs shape-normalised training inputs.
- **Generalisation scope** — splits are identity-disjoint but within the same capture setting; cross-session generalisation is not separately demonstrated.

For the abstract. “On an open-set test of held-out individuals, margin-based face embeddings recognise individual fruit bats above chance (best ROC-AUC ≈ 0.85 – 0.89 ; permutation $p < 0.01$), with the signal shown to be foreground- rather than background-driven — though a silhouette control indicates much of it (nearly all for *Rousettus*) reflects head-crop *shape* rather than facial texture; performance is limited by the number of available identities rather than by model capacity or image resolution, and is not yet robust to background change.”

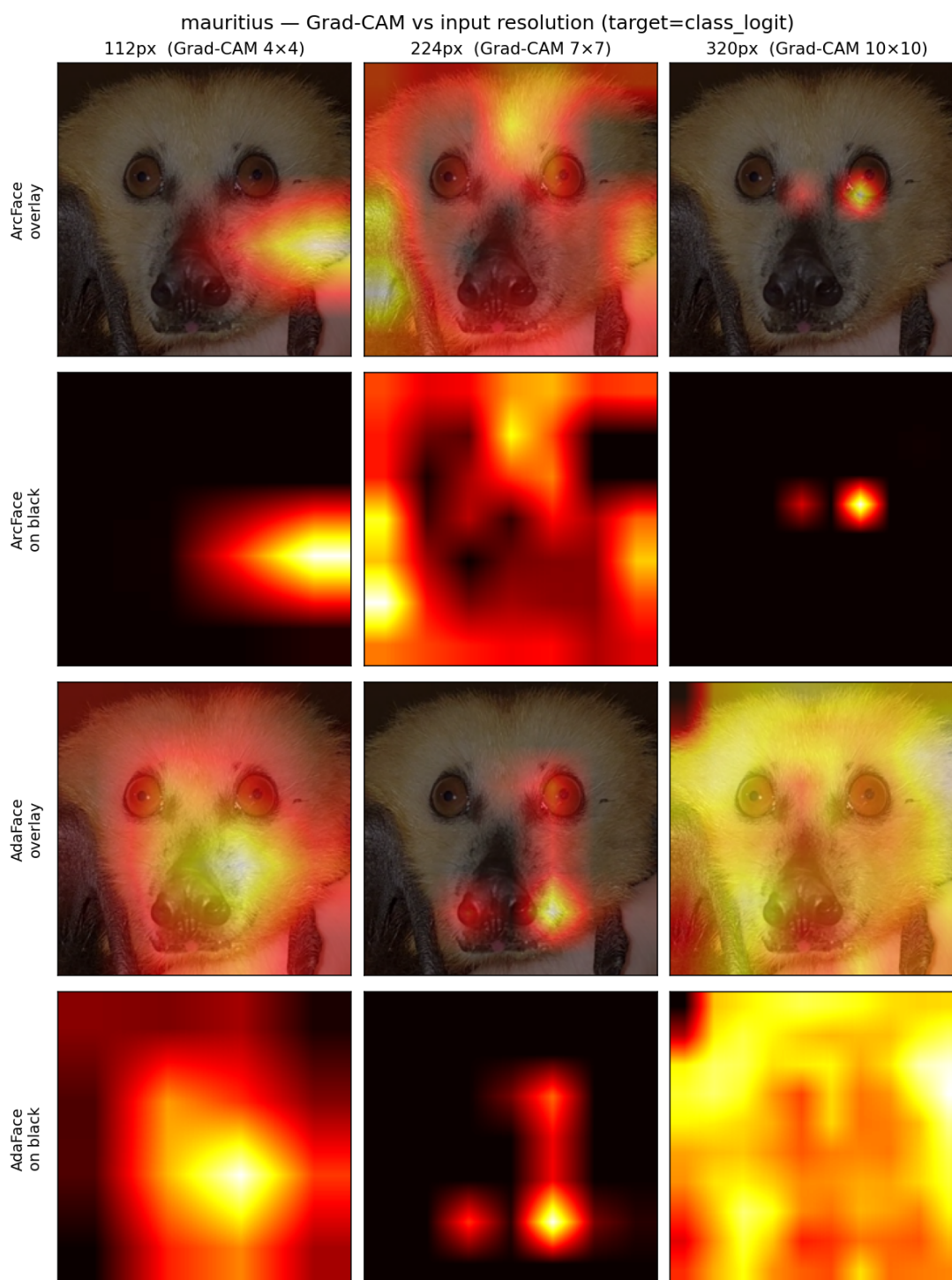


Figure 4. GradCAM (class-logit target) at 112/224/320 px, Mauritius. Attention concentrates on the facial region and sharpens with resolution — yet accuracy is unchanged.



Figure 5. Mauritius — the same test crop as four inputs: natural, background-removed (green), **silhouette** (shape only, zero texture), and **aperture** (texture only, outline removed). The silhouette carries no facial detail, yet is trainable and discriminative.

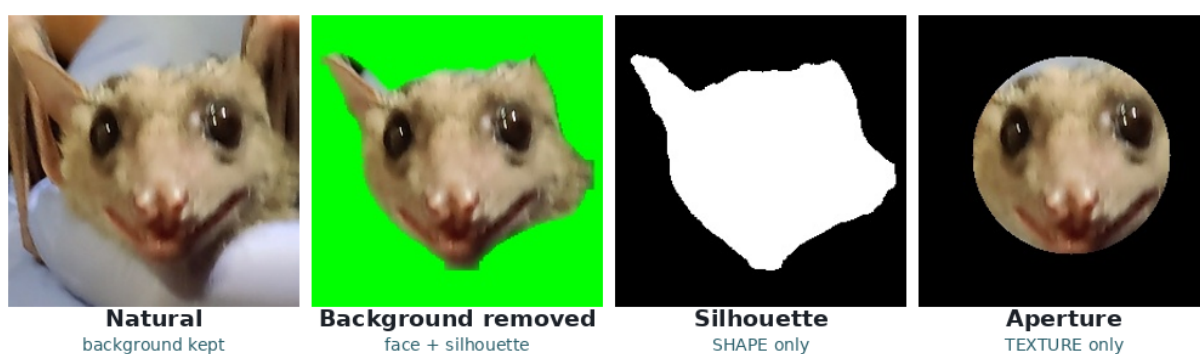


Figure 6. Rousettus — same four inputs. Note how distinctive the bare outline is even with the face erased.